

Guided Project 1

Github: https://github.com/bghosn/geog4057_badrghosn

Summary

This project shows how Python and ArcPy can convert spatial data stored in JSON format into a shapefile for visualization in ArcGIS Pro. The dataset used is the 2018 New Orleans Land Value data downloaded from data.gov. The JSON file contains polygon geometries stored as Well-Known Text (WKT) along with attribute data. Since ArcGIS Pro can't directly import this type of JSON file using standard geoprocessing tools, a custom Python script and Python Toolbox were developed to complete the conversion and visualization.

Code Development

The project was developed using a Python script that reads the JSON file with Python's json library. The script loops through the data records and converts the WKT geometry field into ArcPy polygon geometry using the "arcpy.FromWKT()" function. This conversion is necessary because ArcGIS Pro operates on geometry objects rather than text-based geometry representations.

After converting the geometries, the script creates a new polygon shapefile using "arcpy.management.CreateFeatureclass()". Before making the shapefile, the script checks for an existing output file and deletes it if necessary so that the script can be run multiple times. Attribute fields are then added based on the metadata provided in the JSON file, with field names cleaned and shortened to meet shapefile limitations. Finally, an insert cursor is used to write both the attribute values and polygon geometries into the shapefile.

Interface Design

To make the script for use within ArcGIS Pro, it was integrated into a Python Toolbox. The toolbox allows me to run the script from the geoprocessing pane rather than executing it manually. During the process, I encountered path-related execution errors. These issues were resolved by running the tool in a way that preserved the script's expected folder structure and ensured that the output

workspace was valid. This highlights the importance of understanding how ArcGIS Pro handles workspaces and output paths when creating feature classes programmatically.

How to Use the Code

1. Add the Python Toolbox to ArcGIS Pro.
2. Select the project folder that contains the Python script and the JSON file.
3. Run the tool from the Geoprocessing pane.
4. The output shapefile is created automatically in the specified workspace and can be added to a map for visualization.

Data Visualization

After running the tool, the output shapefile was added to a map in ArcGIS Pro. The polygon features were symbolized using graduated colors for clear visualization. A layout was created that includes a map frame, legend, scale bar, north arrow, and title. The completed layout was exported as a PDF.